LOANAI

AI-Powered Loan Eligibility Platform

AI SPECIALIST - BFSI CAPSTONE PROJECT

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Project domain	Banking, Financial Services and Insurance
Repository	github.com/chetanabonde6-dotcom/ai-loan-checker
Technology	HTML5, CSS3, JavaScript and Google Apps Script
Submission date	26 October 2026

Educational capstone demonstration. Results are illustrative and do not represent a lender's approval decision or financial advice.

1. Executive Summary

LoanAI is a browser-based BFSI platform that brings four common financial planning tools into one responsive interface: a loan eligibility checker, credit-score context, an EMI calculator and an AI-assisted financial tips page.

The project uses transparent calculations and rule-based outcomes. Optional Claude and Google Sheets integrations can extend the demonstration when the user supplies their own configuration, while the main calculators remain usable without external services.

Problem statement

Prospective borrowers often use separate tools to estimate eligibility, interpret a credit score, calculate EMI and understand total repayment. LoanAI combines these tasks and displays the formulas and rules behind its output.

Objectives

- Collect and validate the minimum inputs required for an educational loan assessment.
- Calculate EMI, total repayment and total interest using a reducing-balance formula.
- Classify a user-entered credit score into an educational band.
- Provide understandable reasons and financial guidance.
- Offer optional user-controlled export through Google Apps Script and Sheets.

4	5	0
CORE FINANCIAL TOOLS	MAIN HTML PAGES	BUILD STEPS REQUIRED

2. Functional Modules

Module	Inputs	Output
Loan eligibility	Salary, score, existing EMI, age and profile details	Rule checks, illustrative status and eligible amount
Credit-score analyzer	Score entered by the user	Band, risk context and explanatory feedback
EMI calculator	Principal, annual rate and tenure	Monthly EMI, interest, total payment and amortization
Financial advisor	Financial question and optional API key	Claude response or built-in fallback guidance
Sheets export	Opt-in records and webhook configuration	Eligibility, EMI or chat rows in separate sheet tabs

Typical workflow

1	Open the landing page and select a financial tool
2	Enter fictional or user-provided financial values
3	Validate field ranges before calculation
4	Apply the published rule or financial formula
5	Display the result, reasons and visual explanation
6	Optionally export selected data when Sheets is configured

The tools are educational. A displayed result does not submit a loan application, retrieve a bureau report or guarantee approval.

3. Eligibility and Credit Logic

Loan eligibility rules

The current prototype applies four visible demonstration rules. All four must pass for the illustrative amount to be shown.

Input	Published demonstration rule
Monthly salary	Greater than INR 30,000
User-entered credit score	Greater than 700
Existing monthly EMI	Less than INR 20,000
Applicant age	At least 21 years
Illustrative amount	Monthly salary multiplied by 20 when all rules pass

Credit-score bands

Score range	Band	Educational risk context
750-900	Excellent	Low
650-749	Good	Medium
550-649	Fair	High
300-549	Poor	High

The application analyzes only a score entered by the user. It does not fetch or verify an official bureau score, and its thresholds are not lender underwriting rules.

4. EMI and Repayment Calculations

Reducing-balance EMI

For principal P, monthly interest rate R and tenure N months:

$$EMI = [P \times R \times (1 + R)^N] / [(1 + R)^N - 1]$$

The monthly rate is the annual percentage rate divided by 12 and then by 100. The calculator derives monthly EMI, total repayment and total interest.

Yearly amortization

- Calculate interest on the outstanding balance for each month.
- Subtract monthly interest from EMI to obtain principal repaid.
- Reduce the outstanding balance by the principal portion.
- Aggregate monthly values into a yearly repayment table.

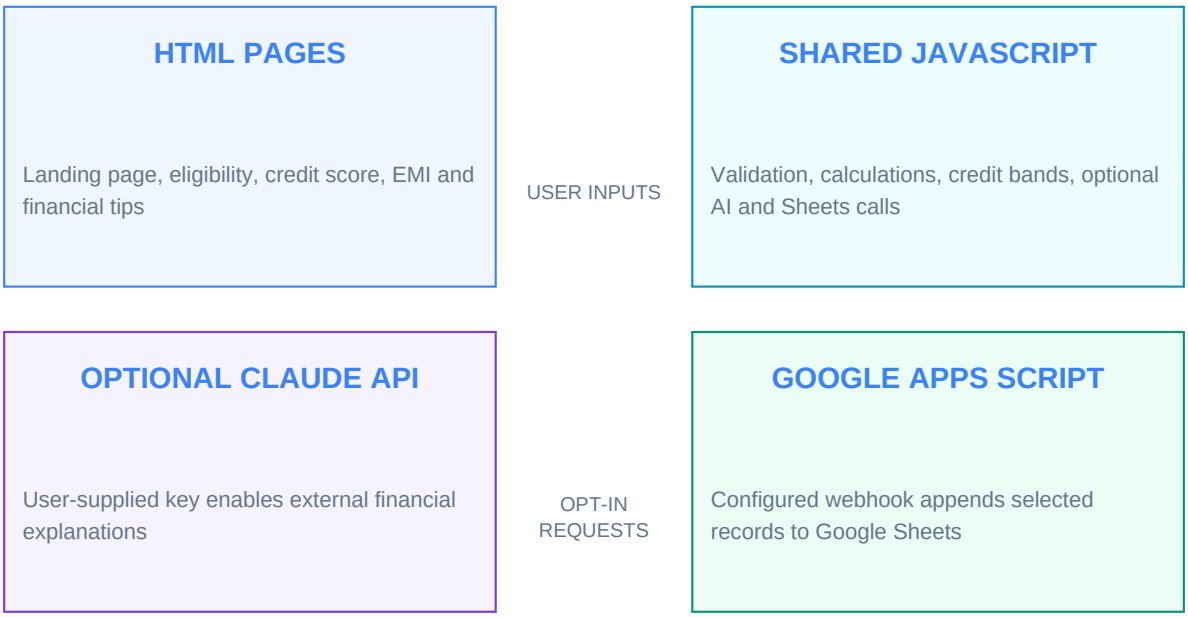
Input validation

Field	Accepted range or rule
Principal	At least INR 10,000
Annual interest rate	Greater than 0% and at most 30%
Tenure	At least one month
Credit score	300-900
Age	18-80 for input validation; eligibility requires 21 or above

The results exclude lender-specific processing fees, taxes, insurance, variable rates and penalty charges.

5. Architecture and Data Flow

LoanAI uses a static browser application with optional external services. The main financial rules and calculations run in JavaScript on the user's device.



Source-code map

Path	Responsibility
index.html	Landing page and module navigation
loan-eligibility.html	Eligibility form, rule results and explanatory feedback
credit-score.html	Credit band visualization and context
emi-calculator.html	EMI inputs, charts and amortization table
ai-tips.html	Financial guidance and integration settings
js/script.js	Shared calculations, validation, AI and Sheets helpers
css/style.css	Responsive glassmorphism design system
google-apps-script/Code.gs	Optional Sheets web-app backend

6. Optional AI and Google Sheets Integration

AI guidance

The financial tips page can call Anthropic's Messages API when the user provides an API key. Without a key, the page uses built-in fallback responses so the demonstration remains functional.

For production use, API calls should be moved behind a secured server endpoint. A browser-stored paid API key can be inspected by the device user and should not be treated as a production secret.

Google Sheets export

The optional Apps Script receives JSON records and places them into separate tabs for eligibility, EMI and chat records. A sheet ID and deployed web-app URL must be supplied by the project owner.

Record type	Example stored fields
Eligibility	Timestamp, name, salary, score, EMI, age, status and illustrative amount
EMI	Principal, rate, tenure, EMI, interest and total amount
Chat	Timestamp, user message, response and session ID

The integration should use fictional demonstration data unless informed consent, access control and an appropriate retention policy are established.

7. Verification, Privacy and Limitations

Verification checklist

Area	Manual verification
Eligibility	Test values around every published threshold
Credit score	Test boundary values 300, 549, 550, 649, 650, 749, 750 and 900
EMI	Compare known loan fixtures and review final amortization balance
Validation	Reject empty, negative and out-of-range inputs
Responsive UI	Review mobile, tablet and desktop layouts
Fallback guidance	Confirm tips remain available without an API key
Sheets	Use only a configured test sheet and fictional records

Privacy

- Calculations run locally unless the user enables an optional integration.
- Claude and Sheets settings are stored in browser localStorage.
- Sheets export can contain financial and identifying fields and therefore requires user control.

Limitations

- Eligibility rules are illustrative and do not represent any lender's policy.
- The application does not retrieve or verify an official credit score.
- Financial tips are educational and not personalized professional advice.
- The repository currently relies on manual verification rather than an automated test suite.

8. Conclusion and Future Scope

LoanAI demonstrates how lightweight web technologies can combine several BFSI learning tools in a single responsive application. Its principal strengths are transparent rules, direct calculations, visual explanations and optional integration paths.

Future enhancements

- Replace browser-direct AI calls with authenticated server-side endpoints.
- Add automated tests for eligibility boundaries, EMI fixtures and amortization.
- Introduce user authentication before any persistent financial-data storage.
- Add rate limiting, audit logs and explicit data-deletion controls.
- Use authorized lender rules or bureau data only with consent and compliance review.

Project links

Resource	Location
GitHub repository	https://github.com/chetanabonde6-dotcom/ai-loan-checker
Main financial logic	js/script.js
Eligibility module	loan-eligibility.html
EMI module	emi-calculator.html
Google Apps Script	google-apps-script/Code.gs

END OF REPORT

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